

Use case for StreamLit as Sale's interface for exploration and analysis of Marketplace data

Introduction

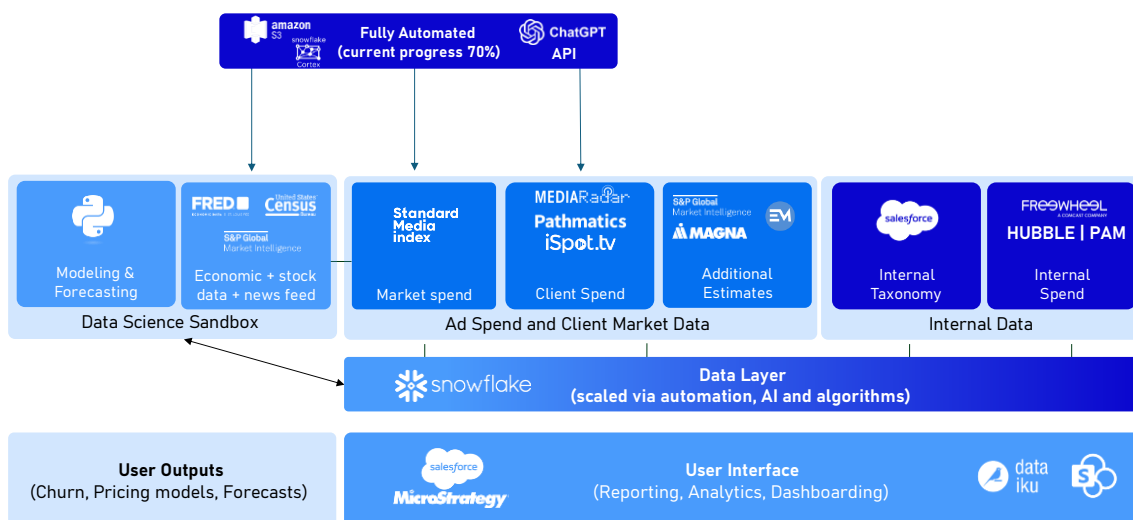
The intent of this use case is to scope the existing limitations of UI tools and provide current progress on building out StreamLit's + Cortex's chatbot capabilities to provide a seamless, easy to use frontend with NLP capabilities to handle route queries from sales such as but not limited to prospecting new clients, a status of spend for existing clients, or a high level overview of competitive activity in the market.

The chatbot currently exists within our current marketplace stack referred to as Project Nexus. Project Nexus aims to fully automate and ingest external data sources about how our advertisers spend in the market, remap and transform the data to make it analogous to our own reporting, and surface it in ways that end users like Sales or Ad Sales Strategy can view the data and make meaningful strategic and tactical decisions based on the reporting and underlying data. This was started to offset thousands of manual hours employees were using to map advertisers from external sources to internal nomenclature and align spend to be in-line with internal data to ensure proper calculations like market share or growth opportunity.

Figure 1 - Nexus Overview

NBCUNIVERSAL

Project Nexus Overview



Current State of the Chatbot

We've progressed to creating a functional chatbot that is able to respond and handle general questions. Additionally, we've created clickable attributes within Streamlit as a proof of concept to facilitate the three basic questions listed above that also tie to our use cases.

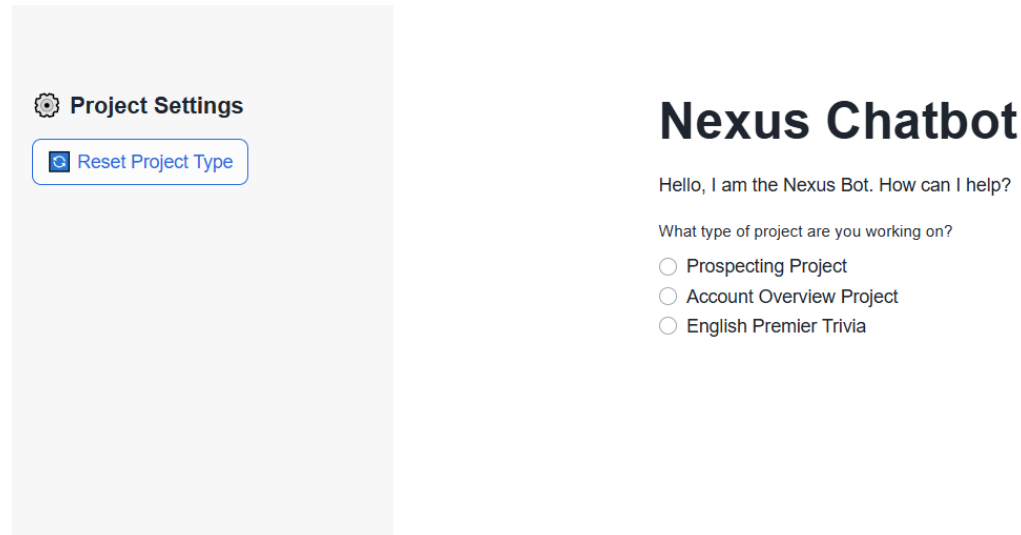


Figure 2 - Current Status of Chatbot

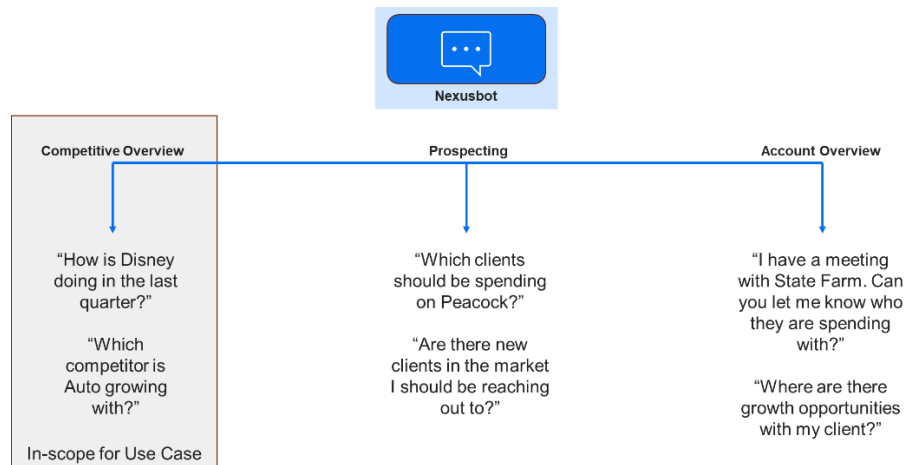
Please note: In this guide, there are frequent references to Marketplace, Market Intelligence, MI, and Marketplace Intelligence. All terms refer to the same team and are various short hands for describing the team responsible for Nexus' creation that sits within the broader Strategy and Insights team within NBCU's Ad Sales division.

Streamlit Use Cases and Current Roadmap

In reviewing current Streamlit documentation, the collective Market Intel team has reached limitations in putting together a basic, functional chatbot that is able to summarize and synthesize quantitative data in our market stack. As new adopters of Streamlit, Cortex and Snowflake overall, several questions have emerged that are potentially simplistic for experience users but remain challenges for our team to successfully scale and deploy our first *usable* chatbot.

For our proposed engagement with Snowflake, we would like to scope assistance in building out just one of the three families of questions listed below as our use case. We believe upon successfully addressing the questions and building the back end successfully around the first group of questions, our team will be sufficiently trained and upskilled to address use cases 2 (prospecting) and 3 (account review). For the sake of clarity, we will include both in the narrative below but are only requesting assistance in scoping use case 1 (competitive overview).

FIGURE 3 – OVERVIEW OF NEXUSBOT USE CASES



The end goal of the chatbot is to successfully handle use cases that fulfill two straight forward criteria: 1- does this question (or family of questions) get asked frequently enough to warrant the time/effort to automate? 2 – Despite the complexity of the phrasing of the question, does the underlying exercise in answering it rely on fairly straight forward queries or fancy filtering to address?

To that end, we provide the following as part of scoping our use cases: what is the objective of the use case?; What are the typical natural language queries that are asked within that use case?; and the existing data tables that are leveraged to address those questions.

1.Competitive Overview [In Scope]

Objective: To properly answer questions about how NBCU’s competitors are pacing in ad revenue for distinct time periods. This includes executive level reporting to better understand how competitors are either successfully or unsuccessfully navigating the advertising market, how they are growing revenue across their defined segments and to help uncover potential opportunity or risk to NBCU’s own revenue growth and business. The deliverable would be in a simple, text based response format that includes both natural language response + summarization of the data into key facts, figures and aggregated metrics.

Typical Queries: “How did Disney pace in Q1?” “Did we grow marketshare against TelevisaUnivision in 2023?” “Can I have a breakout across our competitors to see revenue growth by quarter for the last 5 quarters?”

Datatable: STRATEGY_CLIENT.DEV.MI_MICROSTRATEGY_FACT

2. Prospecting [Out of Scope]

Objective: To successfully provide an end user of a list of either existing clients who have an opportunity to grow revenue on a specific sector [spend with NBCU but not with Peacock] or find completely net new advertisers in the market who don’t spend with NBCU today but should given their historic spend patterns. The deliverable would be in a simple, text based response format that

includes both natural language response + summarization of the data into key facts, figures and aggregated metrics.

Typical Queries: “Who should be my top 5 prospects for Peacock?” “Who should I pitch the Emmy’s to?” “Can I have the top 50 new business prospects to go out and pitch?”

Datatable: STRATEGY_CLIENT.DEV.MI_ADVERTISER_MSTR_FACT

3. Account Review [Out of Scope]

Objective: To provide a basic understanding of how a specific account or media agency is spending in the market with a comparison to how that is similar or different than how said account or agency is spending with NBCU. The deliverable would be in a simple, text based response format that includes both natural language response + summarization of the data into key facts, figures and aggregated metrics.

Typical Queries: “I have a meeting with Nissan next week, can you tell me how they are spending in the market?” “Does my client Amazon spend on sports anywhere else in the market?” “Has PMX shifted dollars to Facebook in the past year?”

Datatable: STRATEGY_CLIENT.DEV.MI_ADVERTISER_MSTR_FACT

Criteria for Success

Upon completion, we would define a successful engagement with the Snowflake team to have accomplished the following items:

- Knowledge and capability to produce serialized sql trees in our current YAML format
- Successfully connect our basic chatbot into our basic semantic model
- Have the chatbot successfully field and respond to a simply natural language query around a competitor (in-scope use case)

Scope of Ask

To date, the Market Intelligence team has successfully created a fully automated ETL process ingesting the market data that is the basis for effectively addressing the above asks. We have also scoped and built a simple, chatbot in Streamlit as well as launched a test POC with Cortex Search Services to index the above scoped data tables. Lastly, we’ve attempted to create a semantic model spec leveraging Snowflake’s user guides for creating both a logical table as well as sample verified queries with SQL trees. However, we’ve reached an impasse and being able to sufficiently connect these tools together to produce a functional chatbot that is able to address at least one of the pre-defined use cases above.

We request Snowflake’s assistance in demonstrating how to connect these functional pieces leveraging out of the box tools from Snowflake (ex. Streamlit, Cortex Analyst) and assist us in

drafting the first round of python to address use case 1. Upon competition, our team will take those key learnings and apply it to use cases 2 and 3 respectively.

To that end, we've also included [as flat files for discussion in email] key artifacts listed below

List of Important Artifacts

- Logical table in YAML format for ingestion into our Semantic model for just the data table STRATEGY_CLIENT.DEV.MI_MICROSTRATEGY_FACT [attached separately]
 - File is named Streamlit_YAML.txt and can also be found in our existing stage for the semantic model listed below
- A sample of verified queries for ingestion into our Semantic model for just the use case 1 [attached separately]
 - File is named Streamlit_queries.txt and can also be found in our existing stage for the semantic model listed below
- Key locations for additional components:
 - Cortex Search Services under DEV.TEST_SEARCH_KEYQUESTIONS
 - Example of a procedure with pseudo code:
STRATEGY_CLIENT.DEV.ACCOUNTREVIEW_PROC
 - Existing stage for semantic model:
STRATEGY_CLIENT/DEV/STREAMLIT_SEMANTIC_MODEL